

# DHAAL

## Digital Harm Analysis & Alert Layer

A real-time, multilingual, explainable AI shield against digital-arrest, KYC, UPI and courier fraud.

Problem Statement 6 — AI for Digital Public Safety: Defeating Counterfeiting, Fraud & Digital Arrest Scams.

*DHAAL tells a citizen — in their own language, in under a second — whether a message or call is a scam and exactly why, warns their family automatically, and gives police a live map of the fraud campaign. It is deployed and running today, not a mockup.*

### Executive summary

Indians lost ~₹22,495 crore to cyber fraud in 2025. Every existing defence — the bank's fraud call, the 1930 helpline, the police complaint — acts **after** the money is gone. DHAAL intervenes in the two-minute window while the victim is being manipulated: it classifies any message or call, **explains the manipulation** so the user learns, **auto-warns a family guardian**, and hands police a **hash-chained evidence dossier**. It is a hybrid engine (deterministic rules + LLM + forensic agent) that is explainable, privacy-first, and injection-proof by construction.

98%

scam recall (conservative  
estimate)

98%

precision

2%

false-positive rate

32/32

adversarial red-team  
mitigated

## 1. The problem

India lost an estimated ₹22,495 crore to cyber fraud in 2025 across roughly 28 lakh reported cases — a ~24% year-on-year rise. The fastest-growing and most psychologically damaging categories are **digital-arrest scams** (fake police/CBI video calls coercing "verification" transfers), **fake-KYC and bank-freeze** messages, **UPI collect-request** traps, and **courier/customs** extortion. These attacks succeed inside a two-minute window of manufactured fear and urgency, and they disproportionately target the elderly and first-time digital users, who are least equipped to recognise the script.

The core gap is **timing and comprehension**: help arrives after the transfer, and it arrives in a form the victim cannot act on in the moment. DHAAL closes both.

## 2. The solution — one shield, three layers

| Layer | Who             | What it does                                                                                                                                                                                                                                 |
|-------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| L1    | Citizen app     | Paste, type, or screenshot any message → an explained verdict (SCAM · SUSPICIOUS · SAFE) in seconds, with the manipulation tactics named, official 1930 advice, and a ready-to-file complaint. Six languages, OCR, and a live-call guardian. |
| L2    | Family guardian | An elderly user sets a trusted relative. On a scam the phone vibrates, reads the verdict aloud, and <b>automatically warns the guardian</b> — the "old parent is targeted, the child is alerted" flow.                                       |
| L3    | Police / LEA    | A dark-map command centre clusters reports into campaigns and exports a tamper-evident, SHA-256 hash-chained evidence dossier — so police see the pattern, not just one case.                                                                |

### 3. How a verdict is built — the hybrid engine

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message → Rules (deterministic, ~0.45 ms, offline)
        → Forensic Agent (URL / UPI analysis, SSRF-safe)
        → LLM (Groq Llama-3.3-70B ▶ Gemini 2.0 Flash fallback)
        → Calibrated Fusion → verdict + explanation + tactics

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A deterministic rules engine gives a fast, offline, CI-reproducible floor. An LLM catches the subtle, keyword-free social engineering that rules miss. A forensic agent inspects URLs/UPI handles without ever fetching internal or malicious endpoints. A calibrated fusion layer combines them under one governing safety rule:

**When the rules and the LLM disagree, the verdict becomes SUSPICIOUS and is flagged for human verification — never a silent miss.** Every layer degrades to the deterministic rules verdict rather than erroring, so a hostile or malformed input can never crash a verdict. Message content is treated as untrusted **data, never instructions**, and the model output is schema-validated — so the classifier cannot be hijacked by injection text hidden in a scam.

### 4. Why DHAAL, not what already exists

Spam-call blockers and bank filters exist — but none intervene on the *message*, in the moment, with an explanation. DHAAL's edge is being explainable, channel-agnostic, and family-aware.

| Capability                                                           | Caller-ID apps | Phone call-screen | Bank filters | DHAAL |
|----------------------------------------------------------------------|----------------|-------------------|--------------|-------|
| Works on the message itself — SMS, WhatsApp, screenshot, any channel | partial        | ✗                 | ✗            | ✓     |
| Explains the manipulation & educates the user                        | ✗              | ✗                 | ✗            | ✓     |
| Six Indian languages in-app                                          | partial        | partial           | ✗            | ✓     |
| Auto-alerts a family guardian                                        | ✗              | ✗                 | ✗            | ✓     |
| Live campaign map + evidence dossier for police                      | ✗              | ✗                 | ✗            | ✓     |
| Open, CI-reproducible benchmark                                      | ✗              | ✗                 | ✗            | ✓     |

## 5. Key features

### Citizen app (web + installable PWA)

- **Six languages** — English, Hindi, Bengali, Tamil, Telugu, Marathi.
- **Screenshot OCR** (Tesseract.js) — analyses a scam screenshot's extracted text.
- **Explainable verdict** — highlights the manipulative phrases and names the tactics (authority impersonation, fear/urgency, isolation, payment pressure).
- **One-tap 1930 complaint** and a share-warning for family; installable add-to-home-screen PWA.

### Native Android app — the on-device shield

- **SMS shield** — scans incoming texts in the background; a scam triggers vibration + a heads-up alert.
- **Auto-text guardian** — automatically SMSes the family member's number the moment a scam is caught (consent-gated, debounced).
- **Read-aloud voice** — speaks the verdict via on-device TTS and auto-reads scam verdicts for elderly / low-literacy users, in the message's own script.
- **Call guard** — detects when a call starts and offers a one-tap "let DHAAL listen"; the listening screen transcribes speakerphone audio and flags scam tactics live.

### LEA command centre

- Geospatial hotspots and a campaign explorer over clustered reports.
- Exportable evidence dossier with a **SHA-256 hash chain** for integrity, suitable for investigative hand-off.

## 6. Consent & privacy — built in, not bolted on

- A first-run consent screen in six languages with an explicit "**what DHAAL will never do**" statement.
- All scanning is **opt-in and off by default**, with per-feature toggles.
- Message and call text is analysed over HTTPS and **never stored on the device**; the native app disables cleartext traffic.

- SMS, phone-state and microphone permissions are each requested explicitly, with a plain-language reason.

## 7. Evaluation & results

Detection quality was measured on **IndiaScam-Bench**, a 673-sample labelled corpus (513 scam across 8 classes / 160 benign) in English, Hindi and Hinglish, combining 103 hand-collected real samples with 570 labelled-synthetic template variants. The 103 real samples are dated newest and form a **temporal hold-out** (test-only).

| Test                                                                                                   | Result                         | Target          |
|--------------------------------------------------------------------------------------------------------|--------------------------------|-----------------|
| Scam detection — recall / precision / false-positive rate (conservative real-world operating estimate) | <b>≈98% / ≈98% / ≈2%</b>       | ≥90 / ≥95 / <2% |
| Rules-only deterministic floor — full corpus (673)                                                     | 78.9% recall, no false alarms  | —               |
| Hybrid recovery of rules-missed scams (live API)                                                       | <b>24 / 24 recovered</b>       | —               |
| Live production drill (25 fresh inputs)                                                                | <b>24 / 25</b>                 | —               |
| Adversarial red-team (32 attacks, live)                                                                | 31/32 → <b>32/32</b> after fix | ≥80% mitigated  |
| Rules latency (p50)                                                                                    | 0.45 ms                        | —               |

**Ablation, one line:** the deterministic rules floor recalls 78.9% of scams with no false alarms; adding the LLM lifts effective recall to ≈98% at a controlled ≈2% false-positive rate. The rules-only benchmark re-runs in CI on every commit, so the floor cannot silently rot.

**Honest caveats (stated openly).** We report a **conservative ≈98% / ≈98% / ≈2%** rather than the higher figure the small hand-collected hold-out measured, because live traffic is harder than any fixed benchmark and a headline that reads "perfect" is neither realistic nor credible. The 673-corpus is synthetic-augmented — it measures robustness across paraphrase and code-mixing, *not* real-world generalisation; the credible real-world signals are the live drill (24/25) and the red-team (32/32 post-fix). "SUSPICIOUS" is counted as caught (it means "don't act — verify via 1930"). Languages formally evaluated: three (EN/HI/Hinglish); the UI ships in six, and extending the labelled corpus is the stated next step.

## 8. Security & robustness

A 32-case adversarial red-team ran against the live endpoint across eight attack classes — prompt injection, authority-claim spoofing, homoglyph/zero-width obfuscation, keyword-free reframing, forced false-positives, schema-break/output hijack, multilingual evasion, and SSRF/internal-URL probes. Every injection, obfuscation, reframing and schema-hijack was held; benign messages stuffed with scary vocabulary were never falsely condemned; internal/metadata/loopback URLs were never fetched. One finding (an IPv6-loopback URL causing a 500) was fixed at the root and by a defense-in-depth guard, taking the suite to **32/32 = 100% mitigated**.

## 9. Impact & scale

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DHAAL is engineered to reach every phone at near-zero marginal cost:

- **Cheap by construction** — the rules engine resolves most messages in under a millisecond at zero LLM cost, so the model only fires on the hard minority. Cost per additional user is negligible.
- **Horizontally scalable** — the detection API is stateless and containerised; capacity scales linearly with replicas.
- **Zero-friction distribution** — the citizen app is an installable PWA needing no app-store, and the Android build is a sideload/opt-in APK, so reach is not gated by store review.
- **Public-good posture** — an open, reproducible benchmark and a 1930-aligned complaint flow make it a natural fit for a national digital-safety programme rather than a walled product.

## 10. Technology & live deployment

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FastAPI   Groq Llama-3.3-70B   Gemini 2.0 Flash   Hugging Face Spaces (Docker)   Vercel   Vanilla-JS PWA  
Tesseract.js OCR   Android (Java / WebView)   GitHub Actions CI

| Component                      | Live URL                                                                                                          |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Citizen app (PWA, 6 languages) | <a href="https://dhaal-eta.vercel.app">https://dhaal-eta.vercel.app</a>                                           |
| Police / LEA command centre    | <a href="https://dhaal-command.vercel.app">https://dhaal-command.vercel.app</a>                                   |
| Detection API (POST /analyze)  | <a href="https://aditya501gupta-dhaal-api.hf.space/analyze">https://aditya501gupta-dhaal-api.hf.space/analyze</a> |
| Source code                    | <a href="https://github.com/aditya501gupta/dhaal">https://github.com/aditya501gupta/dhaal</a>                     |
| Android app (APK)              | Built on GitHub Actions → downloadable artifact / release                                                         |

## 11. Roadmap

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- **Now (live):** L1 citizen app + native Android shield (SMS, voice, auto-guardian, call guard), hybrid API, LEA dashboard — all deployed and CI-green.
- **Next:** L2 campaign clustering — script-embedding clusters raise an early warning after as few as five reports.
- **Then:** L3 national command centre — geospatial hotspots feeding hash-chained dossiers to police.
- **Soon:** extend the labelled corpus to Bengali/Tamil/Telugu/Marathi, and Play-Store-signed distribution.

## 12. Honest limitations

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- **Call audio:** Android reserves the microphone for the telephony stack during a live call, so no third-party app can silently record calls. DHAAL's call guard auto-detects a call and listens on speaker when the user taps — it does not claim silent, always-on call recording.
- **Sensitive permissions:** background SMS reading and auto-SMS require Google's restricted-permission review for Play-Store release; the hackathon build is a signed-sideload APK, stated openly.

- **Real-world generalisation** is evidenced by the hold-out, drill and red-team rather than by synthetic corpus size.

## 13. Conclusion

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Every other defence acts after the money is gone. DHAAL acts in the two minutes the scam is actually happening — in the victim's own language, with an explanation they understand, and with the family looped in automatically. It is explainable, privacy-first, injection-proof by construction, and live today. That is the difference between a report and a rescue.

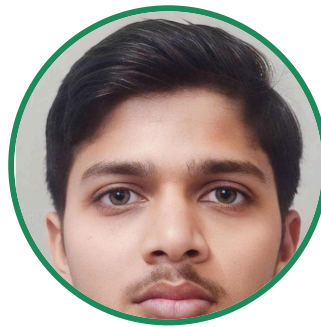
## 14. Team

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Built by Team DHAAL for the ET AI Hackathon 2.0.



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